

tering segments out of the printed sheet and immerse them in water, one at a time. After half a minute or so, depending on the water temperature, you can take the transfer out of the water and feel whether it slides around on the paper. If it does, you can place the transfer where it belongs on the panel. Using tweezers, hold the transfer in the proper place and wipe it carefully with a cotton bud to squeeze out the water underneath the transfer. After the front panel is finished and well dried, it's

a good idea to use apply several thin coats of matt varnish to the surface of the panel (use a spray can for this). Let the front panel dry for half an hour after each coat before applying the next one.

Tips:

1. Use demineralised water if you have hard tap water.
2. Do not use dish detergent to break the surface tension of the water, since it causes soap spots.

If you google "waterslide transfer" or "waterslide decal paper", you will find a lot of information and locations where you can buy it. Some suppliers even sell transfer paper by the sheet, so it's worth looking.

(100387-I)

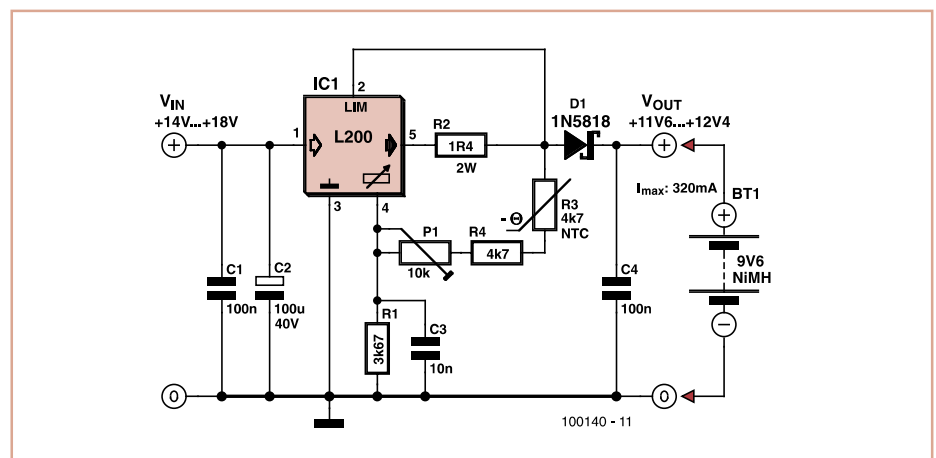
[1] www.papilio.com/laser%20water%20slide%20decal%20paper%20original%20pas.html

L200 charger circuit

Wolfgang Driehaus (Germany)

This circuit came about as the result of an urgent need for a NiMH battery charger. No suitable dedicated IC being immediately to hand, the author pressed an L200 regulator and a 4.7 kΩ NTC thermistor into service. Those components were enough to form the basis of a charger with a cut-off condition based on cell temperature rise rather than relying on the more common negative delta-V detection.

The circuit uses the L200 with the thermistor in the feedback loop. When 'cold' the output voltage of the regulator is about 1.55 V per cell; when 'warm', at a cell temperature of about 35 °C to 40 °C, the output voltage is about 1.45 V per cell and the thermistor has a resistance of about 3.3 kΩ. This temperature sensing is enough to prevent the cells from being overcharged. P1 adjusts the charging voltage, and R2 limits the charge current to 320 mA. The IC is fitted with a small 20 K/W heatsink as it dissipates around 1.2 watts in use.



The charger circuit can be connected permanently to the battery pack. Charging starts when a 'wallwart' adaptor is connected to the input of the charger. The unregulated 12 V supply used by the author delivered an open-circuit voltage of 18 V, dropping to 14 V under load. Even though the charge voltage is reduced when charging

is complete, the cells should not be left permanently on charge.

The author uses the circuit to charge the battery in a torch. After three years and some 150 charge cycles the cells are showing no signs of losing any capacity.

(100140)

AM Receiver with Quadrature Mixer

Gert Baars (The Netherlands)

This circuit is for a superheterodyne receiver where the image frequency is suppressed without the use of an input filter. Instead, it uses two NE(SA)612 type mixer ICs that each work 90° out of phase. With a quadrature front-end, the image frequency is rejected and the noise associated with it disappears. In theory, this increases the sensitivity of the receiver by 6 dB.

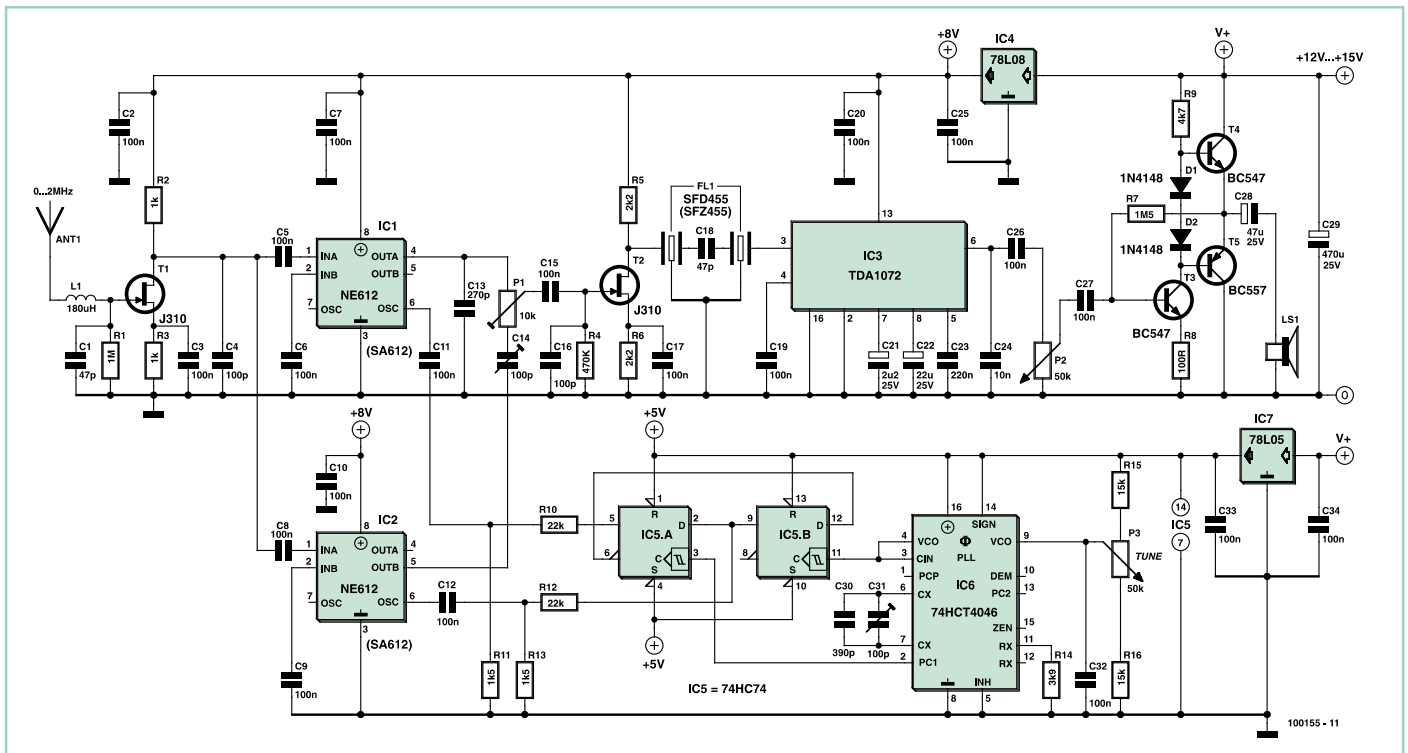
The phase shift of the local oscillator (LO) is provided by two D-type flipflops configured as a ring counter. The outputs of the flipflops always change in the same order. The result

is a frequency that is half that of the oscillator, but with a 90° phase shift between them. These signals are normally called 'Q' (quadrature) and 'I' (in phase).

Phase shifting of the output is provided by two simple RC networks. For the Q mixer the phase shift is set to -45° using a capacitor; for the I mixer it is set to +45° using trimmer (C14). The total phase difference is therefore 90°. The signals are added very simply, using a preset (P1). In this configuration the input frequency is equal to $f_o - f_{if}$ and the image frequency $f_i = f_o + f_{if}$, where the latter is suppressed.

With a low IF, such as used in Software Defined Radio, the phase shifting following the mixers has to cover a relatively wide band, because the IF frequency is low compared to the IF bandwidth. This is much easier to achieve using software rather than a complex phase shifting RC network. With this AM receiver the IF bandwidth is small compared to the centre IF frequency of 455 kHz and the maximum phase error is almost negligible even when a simple RC network is used.

We've used a standard IC for the demodulation: the TDA1072. To drive a loudspeaker



we've added a simple amplifier stage using a pair of transistors (BC547 and BC557) along with a potentiometer (P2) for the volume control.

When setting up the receiver the lowest frequency of the VCO can be configured such that DC can be received. This can be done by ear because the noise disappears and a 50 Hz

hum becomes audible. Setting up the phase shifting can be done with the help of a station that's on the same frequency as the image frequency.

It could happen that the fixed phase shift at the output of the Q mixer isn't exactly -45° , but could be -43° , for example. If you

now adjust the trimmer such that the phase shift becomes $+47^\circ$, the difference becomes 90° again. This is a matter of making small adjustments to the preset and trimmer alternately, while the suppression becomes progressively better until the station is no longer audible due to the image rejection.

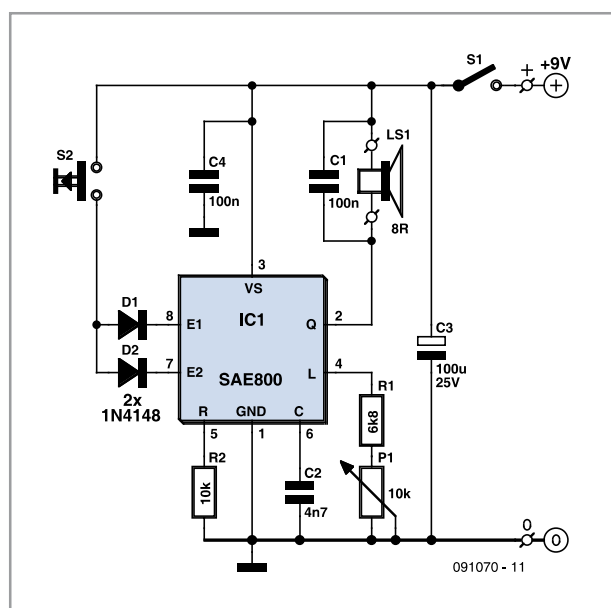
(100155)

Musical Horn for ATBs

Christian Tavernier (France)

If you are both an all-terrain biker and handy with a soldering iron, then we suggest you build this musical horn which, apart from the fact of having a much pleasanter sound than a simple bell, will usually make passers-by turn to you with a broad smile, so surprised will they be to hear these few notes coming from an ATB or mountain-bike.

To achieve this, we have repurposed the SAE800 integrated circuit, which is theoretically designed for doorbells or musical chimes for houses. It takes only a very few external components and can run off any voltage between 2.8 and 18 V. So even with a seriously flat battery, it will go on working — though admittedly at the expense of sound volume. This is relatively high and can be adjusted, to a certain extent,



by potentiometer P1.

Switch S1 is only vital if you want to make the battery last as long as possible. When the circuit is not activated, i.e. when push-button S2 is not pressed, it goes automatically into standby-by mode, when it consumes only a measly $1\mu\text{A}$ or so.

IC1 produces three different sounds, depending on whether E1 or E2, or both together, are activated. This is what we've decided to do here using diodes D1 and D2, as this lets us obtain the most attractive sound, consisting of three notes at 440 Hz, 550 Hz, and 660 Hz, partly overlapping and of decreasing amplitude for around 7 s. Of course, there's nothing to prevent you choosing a different option by fitting only D1 or D2 and leaving the unused input floating.